ARIADNE COLINA-VALERI

ariadnecolina@gmail.com | (813) 847-5642 | https://www.linkedin.com/in/ariadne-colina-valeri

EDUCATION

Tufts University, Medford, MA — Expected Graduation Date: May 2028
Bachelor of Science in Electrical Engineering — GPA: 3.37/4.00
Honors: SHPE Scholar, HSF Scholar, 2024-2026 Dean's List, BEST Scholar, HIFF Scholar, Quest Bridge Scholar
Relevant Coursework: Junior Design, Embedded Systems, Circuits & Electronics, Linear Systems, Modern Physics

PROFESSIONAL EXPERIENCE

GE Transportation / Wabtec, Erie, PA | *Control Systems Engineer Intern* — May 2026 – July 2026
- Automated locomotive diagnostic data extraction by developing Python scripts, reducing manual analysis time
- Built a Fault Analysis Module (FAM) from circuit schematics to isolate fault causes & find failing parts
- Executed failure mode testing by manually triggering faults to validate & refine threshold data feeding the FAM

Tufts Emerging Circuits and Systems Lab, Medford, MA | *Research Assistant* — Sept. 2025 – April 2026
- Applied PINN models to enforce physical validity, keeping generative outputs within realistic switching bounds
- Replaced CVAE with CTGAN to explore beyond conservative data toward higher-performing parameters
- Expanded the training dataset so CTGAN could reliably distinguish top-performing designs from noise

TECHNICAL PROJECTS

Junior Design: Autonomous ESP32 Robot — Sept. 2026 – Present
- Programming a 7-state ESP32 state machine that advances on a button interrupt & takes websocket commands
- Building IR collision avoidance & noise-filtered lane color sensing that drives steering correction
- Designing the robot's physical layout: its outer body in Onshape & a custom ESP32 PCB for its circuits

Embedded Systems: DJ Board on an STM32 Microcontroller — Jan. 2026 – May 2026
- Wrote the state machine & removed audio noise so recordings played back clean, using only on-chip memory
- Validated effect outputs (speed, pitch, etc.) via oscilloscope waveform analysis to confirm correct signal processing
- Implemented seven audio effects (echo, reverse, pitch, speed) enabling users to remix recorded sound in real time

Linear Systems: UAV Pitch Control & RF Signal Restoration — Jan. 2026 – May 2026
- Designed a full PID controller after simpler variants (P, PI, PD) failed to fully stabilize the UAV
- Verified controller stability under real operating conditions & extended the design to cut RF signal recovery error

Digital Logic Circuits: FPGA-Based Audio Synthesizer — Sept. 2025 – Dec. 2025
- Optimized FPGA memory by substituting bit shifts for division to calculate octave frequencies
- Diagnosed chord distortion through wave analysis & resolved it using PWM-based amplitude mixing

ACTIVITIES & LEADERSHIP

Residential Life & Learning | *Residential Assistant* — Jan. 2026 – Present
- Responded to resident incidents & resolved issues in high-pressure situations; mediated disputes & fostered a collaborative living environment in a house of 60 residents

IEEE / SHPE | *Treasurer* — Sept. 2024 – Present
- Managed all purchasing requests through Tufts Treasury, maintained a spending tracker, & rebuilt annual budgets by negotiating for a 10% increase in funding & preventing budget cuts

TECHNICAL SKILLS

- **Programming:** C, C++, Python (PyTorch, pandas, CUDA), MATLAB, SystemVerilog, Arduino, UART
- **Hardware & Design:** AutoCAD, SolidWorks, Onshape
- **Lab Tools & Components:** Oscilloscope, FPGA Board, Analog Discovery 2, STM32 Microcontroller, ESP32

LANGUAGES & INTERESTS

- **Fluent:** English, Spanish, French; Beginner: Italian
- **Interests:** Singing, Rock Climbing, Artistic Swimming (21' USA FL JR. Champion; 22' Olympic Trials Qualifier)